

Moments of RV

1. k-th Moment

For positive integer k ,

kth moment of RV X is $\mathbb{E}[X^k]$

2. Variance of RV

Variance of X is defined as:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}(X))^2]$$

- Captures “spread” of distribution around the mean.
- Standard deviation:

$$\sqrt{\text{Var}(X)}$$

Clearly,

$$\text{Var}(X) \geq 0$$

LEMMA

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}(X))^2$$

PROOF (DISCRETE CASE)

Let $\mu = \mathbb{E}(X)$

$$\text{Var}(X) = \sum_{x \in E_X} p_X(x)(x - \mu)^2$$

Expand:

$$\begin{aligned} &= \sum p_X(x)(x^2 + \mu^2 - 2\mu x) \\ &= \sum p_X(x)x^2 + \sum p_X(x)\mu^2 - 2\mu \sum p_X(x)x \\ &= \mathbb{E}[X^2] + \mu^2 - 2\mu^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \end{aligned}$$

EXAMPLES

a. BERNOULLI

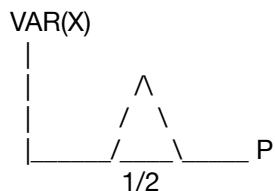
$$X \sim \text{Ber}(p)$$

$$\mathbb{E}[X] = p$$

$$\mathbb{E}[X^2] = p$$

$$\text{Var}(X) = p - p^2 = p(1 - p)$$

Variance as function of



Maximum at $p = \frac{1}{2}$

B. UNIFORM

$$X \sim \text{Unif}(a, b)$$

$$\mathbb{E}(X) = \frac{a + b}{2}$$

$$\mathbb{E}(X^2) = \int_a^b x^2 \frac{1}{b-a} dx = \frac{b^3 - a^3}{3(b-a)} = \frac{b^2 + ab + a^2}{3}$$

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

C. NORMAL

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

$$\text{Var}(X) = \sigma^2$$

4. Expectation of Functions

Random Vector $\bar{X}$, function $g(\bar{X})$

Discrete:

$$\mathbb{E}[g(\bar{X})] = \sum_{\bar{x} \in E_{\bar{X}}} g(\bar{x}) p_{\bar{X}}(\bar{x})$$

Continuous:

$$\mathbb{E}[g(\bar{X})] = \int \cdots \int g(\bar{x}) f_{\bar{X}}(\bar{x}) d\bar{x}$$

For two RVs (X, Y)

Discrete:

$$\mathbb{E}[g(X, Y)] = \sum_x \sum_y g(x, y) p_{XY}(x, y)$$

Continuous:

$$\mathbb{E}[g(X, Y)] = \iint g(x, y) f_{XY}(x, y) dx dy$$

5. Properties of Expectation

1. Linearity

A. $\mathbb{E}[aX] = a\mathbb{E}[X]$

B. $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$

C. Note:

A. Always true — does **NOT** require $X \perp Y$

Proof (Discrete case)

$$\begin{aligned}\mathbb{E}[X + Y] &= \sum_x \sum_y (x + y) p_{XY}(x, y) \\ &= \sum_x \sum_y x p_{XY}(x, y) + \sum_x \sum_y y p_{XY}(x, y) \\ &= \sum_x x p_X(x) + \sum_y y p_Y(y) = \mathbb{E}[X] + \mathbb{E}[Y]\end{aligned}$$

6. Indicator Example

Problem:

Each ball dropped in bin chosen at random.

m balls, n bins.

Find:

$$\mathbb{E}[\# \text{ of empty bins}]$$

Define:

$$X_i = \begin{cases} 1 & \text{if bin } i \text{ empty} \\ 0 & \text{otherwise} \end{cases}$$

$$S = \sum_{i=1}^n X_i$$

Then:

$$\mathbb{E}[X_i] = P(\text{bin } i \text{ empty}) = \left(\frac{n-1}{n}\right)^m$$

$$\mathbb{E}[S] = \sum_{i=1}^n \mathbb{E}[X_i] = n \left(\frac{n-1}{n}\right)^m$$

Independence of X_i 's is **NOT required**

7. Independence

If X, Y independent:

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

Proof (discrete):

$$\mathbb{E}[XY] = \sum_{x,y} xy p_X(x) p_Y(y) = \left(\sum_x x p_X(x) \right) \left(\sum_y y p_Y(y) \right)$$

8. Uncorrelatedness & Covariance

If:

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

Then X, Y are uncorrelated

COVARIANCE

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mu_X\mu_Y$$

If uncorrelated:

$$\text{Cov}(X, Y) = 0$$

If:

$$\mathbb{E}[XY] = 0$$

Then X, Y are **orthogonal**

9. Covariance Matrix

If random vector:

$$X = (X_1, X_2, \dots, X_n)$$

Covariance matrix ($n \times n$) :

$$M_{ij} = \text{Cov}(X_i, X_j)$$

Diagonal elements:

$$M_{ii} = \text{Var}(X_i)$$

10. Important Concept

X, Y independent $\Rightarrow X, Y$ uncorrelated

But converse is **not necessarily true**.

Counter Example

Let:

$$X \in \{-2, -1, 1, 2\} \quad \text{w.p } 1/4 \text{ each}$$

Let:

$$Z = XY$$

Compute:

$$\mathbb{E}[ZX] = \mathbb{E}[X^2Y] = \mathbb{E}[X^2]\mathbb{E}[Y] = 0$$

Since:

$$\mathbb{E}[X] = 0$$

Thus:

$$\mathbb{E}[ZX] = \mathbb{E}[Z]\mathbb{E}[X]$$

So uncorrelated.

But:

$$P(X = 2, Z = 1) \neq P(X = 2)P(Z = 1)$$

Hence not independent.

When does Uncorrelated $\Rightarrow$ Independent?

Only when:

1. Jointly Gaussian RVs
2. Binary RVs

Binary Case Sketch

Let:

$$X \sim \text{Ber}(p), \quad Y \sim \text{Ber}(q)$$

Assume:

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

Then:

$$P(X = 1, Y = 1) = P(X = 1)P(Y = 1)$$

Check other combinations:

$$P(X = 1, Y = 0) = P(X = 1)(1 - P(Y = 1))$$

Similarly for others.

Thus:

$$X \perp Y$$

Big Picture Summary

Concept	Meaning
$\mathbb{E}[X^k]$	kth moment
Variance	Spread around mean
Linearity	Always holds
Independence	Stronger than uncorrelated
Uncorrelated	Cov = 0
Covariance Matrix	Captures pairwise linear relation